

SIDDHANTH JANAWADE

AI Engineer | LLM, RAG & Backend Systems

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SUMMARY

AI and backend engineer (B.Tech Computer Science, 2025) who has built and deployed production RAG pipelines and LLM-backed microservices serving 500+ daily users. Strong in Python and FastAPI, embedding-based retrieval, and measuring model quality with evaluation metrics. Also built an open-source RAG agent with an automated evaluation harness and a lock-free C++20 queue benchmarked at p99.9 latency. Holds four Microsoft Azure AI and data certifications.

SKILLS

AI / ML: LLMs, RAG, Embeddings, Hybrid Search (BM25 + vectors), LangChain, Multi-Agent Systems, Tool Calling, Model Evaluation, PyTorch, TensorFlow, scikit-learn

Languages: Python, C++20, Java, SQL

Backend & Cloud: FastAPI, REST APIs, Docker, Kubernetes, AWS, Azure, CI/CD (GitHub Actions), Git, Production Monitoring

Performance: Concurrency, Lock-Free Data Structures, Latency Benchmarking (p50/p99/p99.9), Profiling

EXPERIENCE

Software Development Engineer, AI & Backend Systems | Hitachi Solutions, Chennai May 2025 – Nov 2025

- Designed, deployed, and maintained a production RAG pipeline with embedding-based retrieval over 50,000+ documents, serving queries for 3 business teams.
- Built and deployed an LLM-backed microservice serving 500+ daily users; added production monitoring for reliability and cut regression resolution time by 60%.
- Ran iterative model evaluation and tuning across 4 sprints, improving retrieval accuracy by 18%.
- Built data pipelines with input validation that catch malformed and edge-case data automatically, cutting downstream errors by 35%.

Technical Intern, Infrastructure & Pre-Sales Engineering | RackBank Datacenters, Indore Jun 2023 – Jul 2023

- Applied AI-based capacity-planning analysis and automated infrastructure workflows.

PROJECTS

DocPilot: RAG Agent with Automated Evaluation | Python, FastAPI, sentence-transformers, Docker, Kubernetes, GitHub Actions github.com/SidJanawade/DocPilot

- Built a RAG agent over technical docs combining BM25 and embedding search with reciprocal rank fusion, a tool-using router (safe calculator, clarification), and a semantic cache that cut average latency from 26.7 ms to 11.7 ms.
- Designed an evaluation harness (Hit@3, MRR, refusal accuracy, grounding) wired into a GitHub Actions quality gate; embeddings reached Hit@3 of 1.00 vs 0.94 for keyword search on a 16-question set that included paraphrased queries.
- Added guardrails that refused 3 of 3 out-of-scope questions; diagnosed over-refusal on paraphrased queries (answer rate 0.88) and documented the fix.
- Containerized the FastAPI service (/ask, /health, /metrics) with Docker and wrote Kubernetes manifests with readiness and liveness probes.

Lock-Free SPSC Queue | C++20, Concurrent Systems

- Designed a lock-free single-producer/single-consumer ring buffer in C++20 using atomic acquire/release ordering and a cache-line-aware layout.
- Built a benchmarking harness measuring p50/p99/p99.9 latency across 1M+ messages, validating the design against a mutex-based baseline; verified zero drops or reordering across 2M messages.

GeNie AI: Multi-Agent Knowledge Assistant | Gemini, LangChain

- Built a multi-agent system with persistent memory and tool calling that tracks tasks and context as connected entities over time.

Image Forgery Detection on CASIA 2.0 | PyTorch, Deep Learning

- Trained a deep-learning classifier in PyTorch reaching 94% detection accuracy, evaluated against real data.

EDUCATION

B.Tech, Computer Science Engineering | SRMIST KTR, Chennai

Aug 2021 – May 2025

CERTIFICATIONS

Microsoft Azure AI Engineer Associate (AI-102) | Data Scientist Associate (DP-100) | Data Engineer Associate (DP-600) | AI Fundamentals (AI-900)